

**ECIL
GET**

**Previous Year Paper
(IT Engg.)
01 Jan, 2026 Shift 2**





इलेक्ट्रॉनिक्स कॉर्पोरेशन ऑफ इंडिया लिमिटेड ELECTRONICS CORPORATION OF INDIA LIMITED

भारत सरकार (परमाणु ऊर्जा विभाग) का उद्यम A Govt. of India (Dept. of Atomic Energy) Enterprise

A Miniratna CPSE



Hall Ticket No	
Candidate Name	
Test Center Name	
Test Date	01/01/2026
Test Time	1:00 PM - 3:00 PM
Subject	CSE or Information Technology

Q.1 A goods train and a superfast train started running from station A and station B, respectively, at the same time towards each other on parallel tracks. If the distance between station A and station B is 260 km and the trains are running with speeds of 40 km/h and 90 km/h, respectively, then at how much distance (in km) from station B will the trains meet?

- Ans
- ☐ 1. 184
 - ☐ 2. 196
 - ☐ 3. 193
 - ☒ 4. 180

Question ID : 441009541200
Option 1 ID : 4410092120199
Option 2 ID : 4410092120198
Option 3 ID : 4410092120197
Option 4 ID : 4410092120196
Chosen Option : 4

Q.2 Identify the antonym of the underlined word in the sentence to fill in the blank.

The speaker's remarks were clearly deliberate, not _____, as some critics have claimed.

- Ans
- ☐ 1. thoughtful
 - ☐ 2. intentional
 - ☐ 3. calculated
 - ☒ 4. spontaneous

Question ID : 4410091124823
Option 1 ID : 4410094434845
Option 2 ID : 4410094434846
Option 3 ID : 4410094434844
Option 4 ID : 4410094434843
Chosen Option : 4

Q.3 Simplify: $40 - [36 - \{48 \div 4 - (6 - 12 \div 2) \div 2\}]$ 6175

Ans  1. 18

 2. 17

 3. 19

 4. 16

Question ID : 441009522459

Option 1 ID : 4410092045257

Option 2 ID : 4410092045255

Option 3 ID : 4410092045258

Option 4 ID : 4410092045256

Chosen Option : 4

Q.4 Identify the appropriate one-word substitute for the following.

A person who interferes in a matter that does not legally concern them

Ans  1. Interloper

 2. Pleader

 3. Contrarian

 4. Revisionist

Question ID : 4410091511572

Option 1 ID : 4410095970353

Option 2 ID : 4410095970354

Option 3 ID : 4410095970355

Option 4 ID : 4410095970356

Chosen Option : 1

Q.5 Complete the following sentence with the correct option.

___ erudite scholar delivered ___ compelling lecture on the intricate nuances of quantum mechanics.

Ans  1. The, a

 2. An, the

 3. The, an

 4. An, a

Question ID : 4410091341557

Option 1 ID : 4410095298314

Option 2 ID : 4410095298315

Option 3 ID : 4410095298317

Option 4 ID : 4410095298316

Chosen Option : 1

Q.6 Which of the following is the correct active voice form of the given sentence?

Every morning, all fresh fruits are delivered by local farmers to the big city market.

- Ans ☒ 1. Every morning, local farmers deliver all fresh fruits to the big city market.
- ☐ 2. Every morning, local farmers have delivered all fresh fruits to the big city market.
- ☐ 3. Every morning, local farmers will deliver all fresh fruits to the big city market.
- ☐ 4. Every morning, all fresh fruits delivered by local farmers to the big city market.

Question ID : 4410091347305

Option 1 ID : 4410095321508

Option 2 ID : 4410095321509

Option 3 ID : 4410095321510

Option 4 ID : 4410095321507

Chosen Option : 2

Q.7 In a certain code language, 'VILE' is coded as '8941' and 'FIVE' is coded as '4219' .
What is the code for 'F' in the given code language?2868

- Ans ☐ 1. 1
- ☐ 2. 9
- ☐ 3. 4
- ☒ 4. 2

Question ID : 441009776870

Option 1 ID : 4410093061485

Option 2 ID : 4410093061486

Option 3 ID : 4410093061483

Option 4 ID : 4410093061484

Chosen Option : 4

Q.8 What should come in place of the question mark (?) in the given series?
131 188 245 302 359 ?

- Ans ☐ 1. 408
- ☐ 2. 420
- ☒ 3. 416
- ☐ 4. 429

Question ID : 4410091499530

Option 1 ID : 4410095923029

Option 2 ID : 4410095923027

Option 3 ID : 4410095923026

Option 4 ID : 4410095923028

Chosen Option : 3

Q.9 Six friends T, U, V, W, Y and Z live on six different floors of the same building. The lowermost floor in the building is numbered 1, the floor above it, number 2 and so on, till the topmost floor is numbered 6. V lives above W and exactly two people live between them. V lives on an even numbered floor but not on floor number 4. Z lives on an odd numbered floor but not on the lowermost floor. Only two people live between Z and T. U lives immediately below Z. Who lives on floor number 4?

Ans  1. U

 2. Z

 3. Y

 4. T

Question ID : 4410091503521

Option 1 ID : 4410095938277

Option 2 ID : 4410095938279

Option 3 ID : 4410095938278

Option 4 ID : 4410095938276

Chosen Option : 1

Q.10 A shopkeeper fixes the marked price of an item 145% above its cost price. What percentage of discount on the marked price should be allowed to gain 47%?

Ans  1. 38%

 2. 40%

 3. 41%

 4. 43%

Question ID : 441009577191

Option 1 ID : 4410092263892

Option 2 ID : 4410092263890

Option 3 ID : 4410092263893

Option 4 ID : 4410092263891

Chosen Option : 2

Q.11 The average of first 13 multiples of 4 is:3839

Ans  1. 13

 2. 4

 3. 26

 4. 28

Question ID : 4410091109976

Option 1 ID : 4410094376900

Option 2 ID : 4410094376899

Option 3 ID : 4410094376901

Option 4 ID : 4410094376898

Chosen Option : 4

Q.12 Fill the blanks using the appropriate option.

The researcher maintained that the new hypothesis could be accepted ___ its underlying assumptions were rigorously tested, ___ the observed anomalies were adequately explained within the revised framework.

- Ans
- ☐ 1. whether ; or
 - ☐ 2. unless ; nor
 - ☒ 3. only if ; and
 - ☐ 4. even though ; yet

Question ID : 4410091511550
Option 1 ID : 4410095970260
Option 2 ID : 4410095970259
Option 3 ID : 4410095970257
Option 4 ID : 4410095970258
Chosen Option : 3

Q.13 The position(s) of how many letters will remain unchanged if each of the letters in the word ABOLISH is arranged in the English alphabetical order?¹⁴⁸⁶

- Ans
- ☐ 1. 1
 - ☐ 2. 3
 - ☐ 3. 4
 - ☒ 4. 2

Question ID : 441009755826
Option 1 ID : 4410092977297
Option 2 ID : 4410092977299
Option 3 ID : 4410092977300
Option 4 ID : 4410092977298
Chosen Option : 4

Q.14 Simplify the expression: $(x + 2)^2 - (x - 3)^2$

- Ans
- ☐ 1. $5x + 10$
 - ☒ 2. $10x - 5$
 - ☐ 3. $5x - 10$
 - ☐ 4. $10x + 5$

Question ID : 441009343787
Option 1 ID : 4410091341135
Option 2 ID : 4410091341134
Option 3 ID : 4410091341137
Option 4 ID : 4410091341136
Chosen Option : 2

Q.15 Select the pair which follows the same pattern as that followed by the two sets of pairs given below. Both pairs follow the same pattern.

JKM-IJL

MNP-LMO8931

- Ans
- ☒ 1. XXY-VWY
 - ☒ 2. XXY-WXZ
 - ☒ 3. XYA-WXZ
 - ☒ 4. XYA-VXY

Question ID : 441009812186

Option 1 ID : 4410093202870

Option 2 ID : 4410093202869

Option 3 ID : 4410093202867

Option 4 ID : 4410093202868

Chosen Option : 3

Q.16 Which of the following statement(s) is/are TRUE about Microcontrollers that are used in Embedded systems?

S1: They need more external hardware

S2: They consume less power

S3: They run only Linux operating systems

S4: They include ROM, RAM, and I/O on-chip

- Ans
- ☒ 1. Only S2
 - ☒ 2. Only S1 and S3
 - ☒ 3. Only S2 and S4
 - ☒ 4. Only S1, S2 and S4

Question ID : 4410091505109

Option 1 ID : 4410095944789

Option 2 ID : 4410095944791

Option 3 ID : 4410095944792

Option 4 ID : 4410095944790

Chosen Option : 3

Q.17 When declaring an array in C, which of the following must be specified explicitly?

- Ans
- ☒ 1. Data type, array name, and the number of elements
 - ☒ 2. Data type and initial values
 - ☒ 3. Array name and memory address
 - ☒ 4. Size and the default values for all elements

Question ID : 4410091529717

Option 1 ID : 4410096041965

Option 2 ID : 4410096041964

Option 3 ID : 4410096041963

Option 4 ID : 4410096041966

Chosen Option : 1

Q.18 When a foreign key uses SET NULL or SET DEFAULT as its referential action, what happens during ON DELETE or ON UPDATE?

- Ans
- ☐ 1. The DBMS prevents deletion or update when either action is used.
 - ☐ 2. The referencing tuple is deleted for SET NULL and replaced for SET DEFAULT.
 - ☒ 3. The referencing attribute is set to NULL for SET NULL and set to its defined default value for SET DEFAULT, for both ON DELETE and ON UPDATE.
 - ☐ 4. SET NULL applies only to ON DELETE, while SET DEFAULT applies only to ON UPDATE.

Question ID : 4410091534943

Option 1 ID : 4410096063018

Option 2 ID : 4410096063017

Option 3 ID : 4410096063015

Option 4 ID : 4410096063016

Chosen Option : 3

Q.19 Which of the following is the best description or an example of Embedded Systems?

- Ans
- ☒ 1. A system dedicated to perform one or a few specific tasks
 - ☐ 2. A general-purpose computer system designed to perform many tasks
 - ☐ 3. A system that cannot be used in real time processing
 - ☐ 4. A system with unlimited memory and processing power

Question ID : 4410091505088

Option 1 ID : 4410095944710

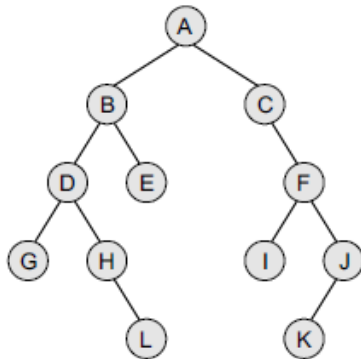
Option 2 ID : 4410095944709

Option 3 ID : 4410095944712

Option 4 ID : 4410095944711

Chosen Option : 1

Q.20 Which of the following gives the correct Preorder and Inorder traversals of the given tree?



- Ans ☒ 1. Preorder: A, B, D, G, H, E, L, C, F, I, J, K Inorder: G, D, H, L, A, B, E, C, I, F, J, K
- ☒ 2. Preorder: A, B, D, G, H, L, E, C, F, I, J, K Inorder: G, D, H, L, B, E, A, C, I, F, K, J
- ☒ 3. Preorder: A, D, G, H, L, B, E, F, I, C, J, K Inorder: D, G, H, B, L, E, A, C, F, I, J, K
- ☒ 4. Preorder: A, B, D, E, C, G, H, L, F, I, J, K Inorder: G, D, L, B, H, E, A, C, I, F, J, K

Question ID : 4410091529806

Option 1 ID : 4410096042325

Option 2 ID : 4410096042323

Option 3 ID : 4410096042324

Option 4 ID : 4410096042326

Chosen Option : 2

Q.21 When inserting a new element at a given position in an array, which operation must be performed before placing the new element?

- Ans ☒ 1. Reverse the array first
- ☒ 2. Delete all elements after the position
- ☒ 3. Shift all elements from the end of the array up to the insertion position one place to the right
- ☒ 4. Shift elements to the left from the start

Question ID : 4410091529757

Option 1 ID : 4410096042114

Option 2 ID : 4410096042112

Option 3 ID : 4410096042111

Option 4 ID : 4410096042113

Chosen Option : 3

Q.22 In SQL, how do CHECK constraints help maintain the consistency of data within a table?

- Ans
- ☐ 1. By automatically generating default values for every attribute in a tuple
 - ☒ 2. By enforcing logical conditions on attribute values so that each tuple inserted or updated remains consistent with defined business rules
 - ☐ 3. By controlling the order in which tuples are inserted into a table
 - ☐ 4. By ensuring foreign keys always reference an existing tuple in another table

Question ID : 4410091534955

Option 1 ID : 4410096063062

Option 2 ID : 4410096063061

Option 3 ID : 4410096063060

Option 4 ID : 4410096063059

Chosen Option : 2

Q.23 What is the average waiting time, using SRTF scheduling, for the following processes?

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	1
P4	5	4

- Ans
- ☐ 1. 4.25 units
 - ☐ 2. 6 units
 - ☒ 3. 3 units
 - ☐ 4. 2.25 units

Question ID : 4410091529874

Option 1 ID : 4410096042601

Option 2 ID : 4410096042602

Option 3 ID : 4410096042600

Option 4 ID : 4410096042599

Chosen Option : 3

Q.24 Which of the following correctly describes a key difference between a traditional page table and an inverted page table?

- Ans
- ☒ 1. A traditional page table has one entry per virtual page, while an inverted page table has one entry per physical frame, making it smaller and independent of virtual address space size.
 - ☐ 2. A traditional page table uses hashing to locate entries, while an inverted page table stores entries sequentially for each virtual page.
 - ☐ 3. An inverted page table requires no chaining because each virtual page maps uniquely to one hash entry.
 - ☐ 4. An inverted page table stores page table entries for each process separately, while a traditional page table combines all processes into one table.

Question ID : 4410091529912

Option 1 ID : 4410096042752

Option 2 ID : 4410096042750

Option 3 ID : 4410096042753

Option 4 ID : 4410096042751

Chosen Option : 1

Q.25 Selection sort requires _____ comparisons for a list of n elements.

Ans

✓ 1. $\frac{n(n-1)}{2}$

✗ 2.

n^2

✗ 3. n

✗ 4. 2n

Question ID : 4410091529836

Option 1 ID : 4410096042439

Option 2 ID : 4410096042440

Option 3 ID : 4410096042441

Option 4 ID : 4410096042442

Chosen Option : 1

Q.26 Which of the following statements are TRUE about the standard protocol MQTT used in IoT?

S1: MQTT is Application layer protocol

S2: MQTT protocol is better suited for battery-powered IoT sensors

S3: MQTT protocol is used ONLY for video streaming

S4: MQTT protocol uses publish-subscribe communication model

Ans

✗ 1. Only S1 and S3

✗ 2. Only S2 and S4

✓ 3. Only S1, S2 and S4

✗ 4. Only S1, S3 and S4

Question ID : 4410091512987

Option 1 ID : 4410095975978

Option 2 ID : 4410095975979

Option 3 ID : 4410095975981

Option 4 ID : 4410095975980

Chosen Option : 3

Q.27 Which of the following describes a weakness of dynamic partitioning?

Ans

✗ 1. Requires dividing memory into equal-size frames

✗ 2. High internal fragmentation

✗ 3. Processes must occupy contiguous partitions

✓ 4. Inefficient processor use due to need for compaction to counter external fragmentation

Question ID : 4410091529896

Option 1 ID : 4410096042687

Option 2 ID : 4410096042688

Option 3 ID : 4410096042686

Option 4 ID : 4410096042689

Chosen Option : 4

Q.28 In a syntax tree used by a compiler, how are operations and their arguments represented?

- Ans
- ☐ 1. Each interior node represents a variable name, while each leaf node represents an operator applied to the variables.
 - ☐ 2. Each interior node represents a lexical token, while its children represent character sequences used to form the token.
 - ☒ 3. Each interior node represents an operation, and its children represent the arguments or operands of that operation.
 - ☐ 4. Each interior node represents an attribute-value pair, while its children represent symbol-table entries.

Question ID : 4410091539686

Option 1 ID : 4410096082078

Option 2 ID : 4410096082077

Option 3 ID : 4410096082080

Option 4 ID : 4410096082079

Chosen Option : 3

Q.29 How can ambiguity be resolved when both base classes contain a display() function?

- Ans
- ☐ 1. By deleting one function in the base class
 - ☒ 2. By explicitly specifying the class name (scope resolution operator)
 - ☐ 3. By overriding both functions in the derived class
 - ☐ 4. By using private inheritance

Question ID : 4410091540403

Option 1 ID : 4410096084919

Option 2 ID : 4410096084921

Option 3 ID : 4410096084920

Option 4 ID : 4410096084922

Chosen Option : 2

Q.30 Suppose we have a relation:
loan(loan-number, branch-name, amount)

Which relational algebra expression retrieves all tuples corresponding to loans greater than ₹12,500 issued by the Agra branch?

- Ans
- ☐ 1. $\sigma_{\text{branch-name} = \text{'Agra'}}(\text{loan}) \div \sigma_{\text{amount} > 12500}(\text{loan})$
 - ☐ 2. $\sigma_{\text{amount} < 12500 \wedge \text{branch-name} = \text{'Agra'}}(\text{loan})$
 - ☐ 3. $\pi_{\text{loan-number}}(\sigma_{\text{amount} > 12500}(\text{loan}))$
 - ☒ 4. $\sigma_{\text{amount} > 12500 \wedge \text{branch-name} = \text{'Agra'}}(\text{loan})$

Question ID : 4410091534867

Option 1 ID : 4410096062717

Option 2 ID : 4410096062715

Option 3 ID : 4410096062718

Option 4 ID : 4410096062716

Chosen Option : 4

Q.31 A temperature sensor in an embedded system typically connects to:

- Ans
- ☐ 1. PWM (Pulse Width Modulation)
 - ☒ 2. ADC (Analog-to-Digital Converter)
 - ☐ 3. DAC (Digital-to-Analog Converter)
 - ☐ 4. UART (Universal Asynchronous Receiver Transmitter)

Question ID : 4410091505139

Option 1 ID : 4410095944911

Option 2 ID : 4410095944910

Option 3 ID : 4410095944912

Option 4 ID : 4410095944909

Chosen Option : 4

Q.32 In the context of the relational model, which of the following statements is TRUE regarding the ordering of tuples?

- Ans
- ☐ 1. Tuples in a relation must appear in the same order as they were inserted to preserve relational integrity.
 - ☒ 2. The order of tuples is irrelevant because a relation is a set, and sets do not impose any ordering on their elements.
 - ☐ 3. Tuples must always be stored in sorted order for a relation to be valid in the relational model.
 - ☐ 4. The order of tuples in a relation affects query results because a relation is defined as an ordered list.

Question ID : 4410091534837

Option 1 ID : 4410096062602

Option 2 ID : 4410096062601

Option 3 ID : 4410096062600

Option 4 ID : 4410096062599

Chosen Option : 2

Q.33 Which of the following correctly distinguishes static RAM (SRAM) from dynamic RAM (DRAM) based on their internal storage mechanisms?

- Ans
- ☐ 1. SRAM needs periodic refresh cycles because capacitor charge leaks, while DRAM does not require any refresh.
 - ☒ 2. SRAM stores each bit using flip-flops and retains data as long as power is supplied, while DRAM stores bits as electric charges in capacitors created by MOS transistors.
 - ☐ 3. Both SRAM and DRAM store bits using capacitors, but SRAM has larger capacitors for longer charge retention.
 - ☐ 4. SRAM stores data as charges in capacitors, whereas DRAM uses flip-flops for long-term retention.

Question ID : 4410091535057

Option 1 ID : 4410096063469

Option 2 ID : 4410096063467

Option 3 ID : 4410096063470

Option 4 ID : 4410096063468

Chosen Option : 2

Q.34 Match the following list regarding the life cycle of Software Development.

List A (Process Activities)	List B (Descriptions)
1. Requirements analysis and definition	A. System is installed, used in practice, and improves over time
2. System and software design	B. Requirements are gathered, understood, and documented
3. Implementation and unit testing	C. Software units are programmed and individual specifications
4. Integration and system testing	D. Requirements are partitioned into hardware/software architecture is defined
5. Operation and maintenance	E. Units are combined and evaluated to check conformance with overall requirements

- Ans ☐ 1. 1-C, 2-B, 3-A, 4-D, 5-E
☒ 2. 1-B, 2-D, 3-C, 4-E, 5-A
☐ 3. 1-E, 2-A, 3-B, 4-C, 5-D
☐ 4. 1-A, 2-C, 3-D, 4-B, 5-E

Question ID : 4410091539754
Option 1 ID : 4410096082345
Option 2 ID : 4410096082348
Option 3 ID : 4410096082346
Option 4 ID : 4410096082347
Chosen Option : 2

Q.35 In a linear queue implemented using an array, what does the condition FRONT = -1 and REAR = -1 indicate?

- Ans ☐ 1. The queue is in overflow state.
☒ 2. The queue is empty.
☐ 3. The queue is full.
☐ 4. Only one element is present in the queue.

Question ID : 4410091529786
Option 1 ID : 4410096042238
Option 2 ID : 4410096042237
Option 3 ID : 4410096042235
Option 4 ID : 4410096042236
Chosen Option : 2

Q.36 The while loop is sometimes called an entry-control loop because:

- Ans ☐ 1. It guarantees that the statements within the loop will execute at least once.
☐ 2. It requires a semicolon after the while (test condition) line.
☒ 3. It controls the execution of the loop's body before the statements inside are executed.
☐ 4. It is the first loop type taught in programming courses.

Question ID : 4410091540336
Option 1 ID : 4410096084658
Option 2 ID : 4410096084656
Option 3 ID : 4410096084657
Option 4 ID : 4410096084655
Chosen Option : 3

Q.37 Which of the following statements about Breadth-First Search (BFS) is correct?

- Ans
- ☒ 1. BFS is neither complete nor optimal because it may miss shorter paths in an unweighted graph.
 - ☒ 2. BFS is complete only for finite graphs and optimal for all weighted and unweighted graphs.
 - ☒ 3. BFS is complete even for infinite graphs because it explores in layers and eventually reaches every node.
 - ☒ 4. BFS is complete for any graph where a solution exists, but it is optimal only when all edges have equal cost.

Question ID : 4410091529830

Option 1 ID : 4410096042418

Option 2 ID : 4410096042416

Option 3 ID : 4410096042417

Option 4 ID : 4410096042415

Chosen Option : 4

Q.38 When multiple objects of a class access a static variable, how is its value handled?

- Ans
- ☒ 1. Each object receives its own separate copy, and changes made by one object do not affect others.
 - ☒ 2. The static variable can be accessed only after creating all objects of the class.
 - ☒ 3. There is only one shared copy of the static variable, and any update by one object affects the shared value.
 - ☒ 4. The static variable is initialized every time a new object is created.

Question ID : 4410091540398

Option 1 ID : 4410096084899

Option 2 ID : 4410096084902

Option 3 ID : 4410096084901

Option 4 ID : 4410096084900

Chosen Option : 3

Q.39 For the relation loan-officer (customer-name, banker-name, loan-number), what relational algebra expression retrieves all customers whose customer-name is identical to the banker-name?

- Ans
- ☒ 1. $\sigma_{\text{customer-name} = \text{banker-name}}(\text{loan-officer})$
 - ☒ 2. $\pi_{\text{loan-number}}(\sigma_{\text{customer-name} = \text{banker-name}}(\text{loan-officer}))$
 - ☒ 3. $\sigma_{\text{customer-name} \wedge \text{banker-name}}(\text{loan-officer})$
 - ☒ 4. $\pi_{\text{customer-name}}(\sigma_{\text{banker-name} = \text{loan-number}}(\text{loan-officer}))$

Question ID : 4410091534904

Option 1 ID : 4410096062856

Option 2 ID : 4410096062858

Option 3 ID : 4410096062855

Option 4 ID : 4410096062857

Chosen Option : 1

Q.40 Which of the following is NOT an unsupervised learning algorithm?

- Ans
- ☐ 1. PCA (Principal Component Analysis)
 - ☐ 2. LLE (Locally Linear Embedding)
 - ☐ 3. t-SNE (t-distributed Stochastic Neighbor Embedding)
 - ☒ 4. Logistic Regression

Question ID : 4410091539738

Option 1 ID : 4410096082283

Option 2 ID : 4410096082281

Option 3 ID : 4410096082282

Option 4 ID : 4410096082284

Chosen Option : 4

Q.41 `float x = 3.14;`
`void *vptr = &x;`
Which is a valid way to access the value of x through vptr?

- Ans
- ☐ 1. `printf("%f", *(double*)vptr);`
 - ☒ 2. `printf("%f", *(float*)vptr);`
 - ☐ 3. `printf("%f", *vptr);`
 - ☐ 4. `printf("%d", *(int*)vptr);`

Question ID : 4410091540358

Option 1 ID : 4410096084746

Option 2 ID : 4410096084743

Option 3 ID : 4410096084744

Option 4 ID : 4410096084745

Chosen Option : 2

Q.42 What happens when a DELETE statement is executed without a WHERE clause?

- Ans
- ☐ 1. No tuple is deleted.
 - ☒ 2. All tuples in the table are deleted, but the table structure remains.
 - ☐ 3. Only one (first) tuple is deleted from the relation.
 - ☐ 4. The entire table is removed from the database.

Question ID : 4410091534935

Option 1 ID : 4410096062978

Option 2 ID : 4410096062975

Option 3 ID : 4410096062977

Option 4 ID : 4410096062976

Chosen Option : 2

Q.43 Which protocol ensures reliable data transmission in IoT applications?

- Ans ☒ 1. TCP
☐ 2. ICMP
☐ 3. ARP
☐ 4. UDP

Question ID : 4410091512982
Option 1 ID : 4410095975959
Option 2 ID : 4410095975960
Option 3 ID : 4410095975961
Option 4 ID : 4410095975958
Chosen Option : 1

Q.44 Match the components in List A with their respective functions in List B.

List A (Components)	List B (Functions)
1. Compiler	A. Executes the source program statement by statement
2. Interpreter	B. Produces machine-language target program from a high-level source
3. Assembler	C. Converts assembly language into machine code
4. Linker	D. Combines object files and libraries into a single executable

- Ans ☐ 1. 1-D, 2-B, 3-A, 4-C
☐ 2. 1-C, 2-D, 3-A, 4-B
☐ 3. 1-A, 2-C, 3-B, 4-D
☒ 4. 1-B, 2-A, 3-C, 4-D

Question ID : 4410091539676
Option 1 ID : 4410096082038
Option 2 ID : 4410096082040
Option 3 ID : 4410096082037
Option 4 ID : 4410096082039
Chosen Option : 4

Q.45 Which operator performs a bit-by-bit inversion (flips 0 to 1 and 1 to 0)?

- Ans ☐ 1. ^
☒ 2. ~
☐ 3. >>
☐ 4. <<

Question ID : 4410091540314
Option 1 ID : 4410096084575
Option 2 ID : 4410096084576
Option 3 ID : 4410096084577
Option 4 ID : 4410096084578
Chosen Option : 2

Q.46 Which object is used for standard input in C++?

- Ans
- ☐ 1. clog
 - ☒ 2. cin
 - ☐ 3. cout
 - ☐ 4. cerr

Question ID : 4410091540375
Option 1 ID : 4410096084814
Option 2 ID : 4410096084813
Option 3 ID : 4410096084811
Option 4 ID : 4410096084812
Chosen Option : 2

Q.47 Which of the following statements are TRUE about standard 8051 microcontroller?

S1: It has 16 general purpose registers
S2: The program memory space is 1 Megabyte
S3: It has 2 timers (timer0 and timer1)
S4: It has 4 ports (p0, p1, p2, p3)

- Ans
- ☐ 1. Only S1, S2 and S4
 - ☐ 2. Only S1 and S3
 - ☐ 3. Only S2 and S4
 - ☒ 4. Only S3 and S4

Question ID : 4410091512966
Option 1 ID : 4410095975897
Option 2 ID : 4410095975894
Option 3 ID : 4410095975895
Option 4 ID : 4410095975896
Chosen Option : 1

Q.48 What major problem can occur in preemptive scheduling when two processes share data?

- Ans
- ☒ 1. A race condition may occur because one process may be preempted while updating shared data, leaving the data in an inconsistent state.
 - ☐ 2. Deadlock always occurs when a process is preempted.
 - ☐ 3. The operating system automatically blocks all shared-memory operations.
 - ☐ 4. Context switching becomes impossible.

Question ID : 4410091529862
Option 1 ID : 4410096042545
Option 2 ID : 4410096042543
Option 3 ID : 4410096042546
Option 4 ID : 4410096042544
Chosen Option : 1

Q.49 The signal lines in a system bus are generally organized into which three functional groups?

- Ans
- ☒ 1. Interrupt lines, DMA lines, and cache coherence lines
 - ☒ 2. Data lines, address lines, and control lines
 - ☒ 3. Arithmetic lines, logic lines, and timing lines
 - ☒ 4. Input lines, output lines, and bidirectional lines

Question ID : 4410091535042
Option 1 ID : 4410096063409
Option 2 ID : 4410096063408
Option 3 ID : 4410096063407
Option 4 ID : 4410096063410
Chosen Option : 2

Q.50 When using the SQL DROP command, what is the correct interpretation of the CASCADE and RESTRICT options?

- Ans
- ☒ 1. CASCADE automatically drops the specified schema element along with all objects (tables, views, constraints) that depend on it, whereas RESTRICT drops it only if no dependent objects exist.
 - ☒ 2. CASCADE drops the specified schema element only if it contains no tables, while RESTRICT removes all dependent objects automatically.
 - ☒ 3. RESTRICT drops the specified object even if it is referenced elsewhere, while CASCADE prevents the object from being dropped if dependencies exist.
 - ☒ 4. CASCADE removes only the data but retains the table definition, while RESTRICT deletes both data and definition.

Question ID : 4410091534927
Option 1 ID : 4410096062944
Option 2 ID : 4410096062943
Option 3 ID : 4410096062946
Option 4 ID : 4410096062945
Chosen Option : 1

Q.51 Consider the following Grammar:

$E \rightarrow TE'$
 $E' \rightarrow +TE' \mid \epsilon$
 $T \rightarrow FT'$
 $T' \rightarrow *FT' \mid \epsilon$
 $F \rightarrow (E)id$

Which of the following statements is true?

- Ans
- ☒ 1. Since E' derives ϵ , FOLLOW(E') must be empty, because nothing follows E' in its productions.
 - ☒ 2. FIRST(E) contains + because E' begins with + in one of its productions.
 - ☒ 3. FOLLOW(T) contains) because T appears before E' in $E \rightarrow TE'$ and E' can derive ϵ , so FOLLOW(E) is added to FOLLOW(T).
 - ☒ 4. FOLLOW(F) does not include \$ because F never appears at the end of any production.

Question ID : 4410091539698
Option 1 ID : 4410096082125
Option 2 ID : 4410096082126
Option 3 ID : 4410096082128
Option 4 ID : 4410096082127
Chosen Option : 3

Q.52 Which of the following correctly identifies one of the four fundamental activities that must appear in every software process?

- Ans ☒ 1. Software validation – ensuring the software actually meets customer requirements
- ☐ 2. Software deployment – delivering the executable files to end users
- ☐ 3. Software risk analysis – highlighting the project risks
- ☐ 4. Software documentation – preparing manuals and installation guides

Question ID : 4410091539766
Option 1 ID : 4410096082401
Option 2 ID : 4410096082402
Option 3 ID : 4410096082404
Option 4 ID : 4410096082403
Chosen Option : 4

Q.53 In a binary search algorithm applied on an array of n elements, the total number of comparisons required to locate an element is approximately:

- Ans ☐ 1. $\sqrt{n}$
- ☒ 2. $\log_2 n$
- ☐ 3. n
- ☐ 4. n/2

Question ID : 4410091529759
Option 1 ID : 4410096042122
Option 2 ID : 4410096042121
Option 3 ID : 4410096042119
Option 4 ID : 4410096042120
Chosen Option : 2

Q.54 Which of the following is TRUE about an Embedded System?

- Ans ☐ 1. The system always has only CPU and I/O devices
- ☐ 2. The system always contains only CPU
- ☐ 3. The system always has only CPU and an operating system
- ☒ 4. The system has CPU, memory, I/O devices, and application software

Question ID : 4410091505100
Option 1 ID : 4410095944755
Option 2 ID : 4410095944753
Option 3 ID : 4410095944754
Option 4 ID : 4410095944756
Chosen Option : 4

Q.55 If there are n processes in the ready queue and the time quantum is q , what is the maximum time a process may have to wait before receiving its next time slice?

- Ans
- ☐ 1. $q \times (n + 1)$
 - ☒ 2. $(n - 1) \times q$
 - ☐ 3. q / n
 - ☐ 4. $n \times q$

Question ID : 4410091529878
Option 1 ID : 4410096042617
Option 2 ID : 4410096042618
Option 3 ID : 4410096042616
Option 4 ID : 4410096042615
Chosen Option : 2

Q.56 In a bus system that uses polling instead of a traditional bus-grant signal, how is access to the bus typically controlled?

- Ans
- ☐ 1. The bus controller broadcasts a single interrupt signal, and devices compete using priority encoding.
 - ☐ 2. Devices autonomously take control of the bus once they detect that the bus-busy line is low, without any intervention from a controller.
 - ☐ 3. Each device asserts a dedicated grant line to request bus ownership directly from the controller.
 - ☒ 4. The bus controller cycles through programmable poll lines that represent device addresses; when a device recognizes its address, it asserts the bus-busy line and gains bus access.

Question ID : 4410091535086
Option 1 ID : 4410096063592
Option 2 ID : 4410096063594
Option 3 ID : 4410096063591
Option 4 ID : 4410096063593
Chosen Option : 4

Q.57 The overloaded extraction operator ($>>$) is associated with which class?

- Ans
- ☐ 1. `istream`
 - ☐ 2. `ostream`
 - ☒ 3. `istream`
 - ☐ 4. `streambuf`

Question ID : 4410091540409
Option 1 ID : 4410096084946
Option 2 ID : 4410096084943
Option 3 ID : 4410096084944
Option 4 ID : 4410096084945
Chosen Option : 1

Q.58 Which of the following correctly describes the behavior of POP3 in its two operating modes?

- Ans
- ☐ 1. In delete mode, messages remain on the server; in keep mode, messages are removed automatically after each retrieval.
 - ☐ 2. In both delete and keep modes, POP3 always deletes the mail after it is flagged as read.
 - ☐ 3. Delete mode is used only when accessing mail remotely; keep mode is used only on a permanent computer.
 - ☒ 4. In delete mode, messages are removed from the mailbox after retrieval; in keep mode, messages remain in the mailbox even after being downloaded.

Question ID : 4410091535012

Option 1 ID : 4410096063287

Option 2 ID : 4410096063289

Option 3 ID : 4410096063288

Option 4 ID : 4410096063290

Chosen Option : 4

Q.59 Which of the following is an assumption in logistic regression?

- Ans
- ☐ 1. The dependent and independent variables must have a strictly linear relationship.
 - ☒ 2. The independent variables need to be linearly related to the log-odds (logit) rather than directly to the dependent variable.
 - ☐ 3. The dependent variable must be continuous, and the independent variables must be categorical.
 - ☐ 4. The independent variables are required to follow a normal distribution.

Question ID : 4410091539722

Option 1 ID : 4410096082219

Option 2 ID : 4410096082218

Option 3 ID : 4410096082220

Option 4 ID : 4410096082217

Chosen Option : 2

Q.60 Which memory is used for storing variables during execution in Embedded Systems?

- Ans
- ☒ 1. RAM
 - ☐ 2. EPROM
 - ☐ 3. FLASH
 - ☐ 4. ROM

Question ID : 4410091505148

Option 1 ID : 4410095944947

Option 2 ID : 4410095944948

Option 3 ID : 4410095944946

Option 4 ID : 4410095944945

Chosen Option : 1

Q.61 What will happen if a constructor is declared with a return type?

- Ans
- ☒ 1. Compiler error
 - ☒ 2. Treated as a normal function
 - ☒ 3. Constructor becomes overloaded
 - ☒ 4. Returned value is ignored

Question ID : 4410091540388

Option 1 ID : 4410096084859

Option 2 ID : 4410096084860

Option 3 ID : 4410096084861

Option 4 ID : 4410096084862

Chosen Option : 3

Q.62 Which of the following is a correct function definition (returning nothing and taking no parameters)?

- Ans
- ☒ 1. void f(int) { /* ... */ }
 - ☒ 2. void f(0); { /* ... */ }
 - ☒ 3. void f(void) { /* ... */ }
 - ☒ 4. f(void) void { /* ... */ }

Question ID : 4410091540347

Option 1 ID : 4410096084700

Option 2 ID : 4410096084702

Option 3 ID : 4410096084699

Option 4 ID : 4410096084701

Chosen Option : 3

Q.63 What will be the output of the following code?

```
int a = 5, b = 10;
if(a = b)
printf("True");
else
printf("False");
```

- Ans
- ☒ 1. FALSE
 - ☒ 2. TRUE
 - ☒ 3. No Output
 - ☒ 4. Compiler Error

Question ID : 4410091540329

Option 1 ID : 4410096084628

Option 2 ID : 4410096084627

Option 3 ID : 4410096084630

Option 4 ID : 4410096084629

Chosen Option : 2

Q.64 Match List A with List B (Infix to Postfix Conversion Algorithm).

List A	List B
1. "(" is encountered	A. Add the operand to the postfix expres
2. An operand is encountered	B. Push it onto the stack
3. ")" is encountered	C. Pop from stack to postfix until "(" is fc "("
4. An operator 0 is encountered	D. Pop operators of same or higher preci

- Ans
- ☒ 1. A. 1-A, 2-B, 3-C, 4-D
 - ☒ 2. C. 1-C, 2-A, 3-B, 4-D
 - ☒ 3. D. 1-B, 2-D, 3-A, 4-C
 - ☒ 4. B. 1-B, 2-A, 3-C, 4-D

Question ID : 4410091529780
Option 1 ID : 4410096042211
Option 2 ID : 4410096042213
Option 3 ID : 4410096042214
Option 4 ID : 4410096042212
Chosen Option : 4

Q.65 Why does a class not create memory space by itself?

- Ans
- ☒ 1. Because a class is only a specification/template
 - ☒ 2. Because memory must be manually allocated
 - ☒ 3. Because only structs can create memory
 - ☒ 4. Because C++ disallows static allocation

Question ID : 4410091540392
Option 1 ID : 4410096084876
Option 2 ID : 4410096084877
Option 3 ID : 4410096084878
Option 4 ID : 4410096084875
Chosen Option : 1

Q.66 Which feature of OOP directly enables 'plug-and-play' software reusability?

- Ans
- ☒ 1. Inheritance
 - ☒ 2. Encapsulation
 - ☒ 3. Dynamic binding
 - ☒ 4. Abstraction

Question ID : 4410091540373
Option 1 ID : 4410096084807
Option 2 ID : 4410096084808
Option 3 ID : 4410096084810
Option 4 ID : 4410096084809
Chosen Option : 1

Q.67 Which of the following correctly describes the roles of the write head and read head in a magnetic disk system?

- Ans
- ☐ 1. The write head and read head perform identical functions because both detect magnetic transitions during rotation.
 - ☐ 2. The write head senses magnetic field changes from the rotating disk, while the read head generates magnetic spots on the surface.
 - ☐ 3. The write head and read head both remain stationary, but neither interacts directly with magnetic changes on the surface.
 - ☒ 4. The write head records bits as magnetic spots on the rotating disk surface, while the read head detects stored bits by sensing changes in magnetic fields as the surface passes beneath it.

Question ID : 4410091535078

Option 1 ID : 4410096063552

Option 2 ID : 4410096063554

Option 3 ID : 4410096063553

Option 4 ID : 4410096063551

Chosen Option : 4

Q.68 Which of the following statements is TRUE about the UART communication protocol in an embedded system?

- Ans
- ☒ 1. It is a serial and asynchronous communication protocol
 - ☐ 2. It is a serial and synchronous communication protocol
 - ☐ 3. It is a parallel and asynchronous communication protocol
 - ☐ 4. It is a parallel and synchronous communication protocol

Question ID : 4410091507678

Option 1 ID : 4410095955316

Option 2 ID : 4410095955315

Option 3 ID : 4410095955314

Option 4 ID : 4410095955313

Chosen Option : 1

Q.69 Which of the following best describes the purpose of a CHECK constraint in SQL?

- Ans
- ☐ 1. It ensures that tuples are stored in sorted order according to a specified attribute.
 - ☐ 2. It enforces referential integrity by automatically matching foreign keys to primary keys.
 - ☐ 3. It prevents deletion of tuples that violate the NOT NULL constraint.
 - ☒ 4. It ensures the validity of data by restricting attribute values so that each inserted or updated tuple satisfies a specified condition.

Question ID : 4410091534946

Option 1 ID : 4410096063027

Option 2 ID : 4410096063029

Option 3 ID : 4410096063030

Option 4 ID : 4410096063028

Chosen Option : 4

Q.70 What two data types are explicitly stated as valid for a Switch Case Value?

- Ans
- ☐ 1. Float type and int type
 - ☐ 2. Only float type
 - ☒ 3. Character type and int type
 - ☐ 4. Character type

Question ID : 4410091540334
Option 1 ID : 4410096084648
Option 2 ID : 4410096084650
Option 3 ID : 4410096084647
Option 4 ID : 4410096084649
Chosen Option : 3

Q.71 Which of the following statement(s) is/are TRUE about GPIO digital pins on a microcontroller?

S1: GPIO pins can be configured for reading signal from sensor
S2: GPIO pins can be configured for sending output to external devices such as LED
S3: GPIO pins can be used ONLY for interrupt handling

- Ans
- ☐ 1. Only S1 and S3
 - ☐ 2. Only S2 and S3
 - ☒ 3. Only S1 and S2
 - ☐ 4. Only S1

Question ID : 4410091507669
Option 1 ID : 4410095955279
Option 2 ID : 4410095955280
Option 3 ID : 4410095955278
Option 4 ID : 4410095955277
Chosen Option : 3

Q.72 What is the default storage class for a variable declared inside a function if no storage specifier is given?

- Ans
- ☐ 1. static
 - ☒ 2. auto
 - ☐ 3. extern
 - ☐ 4. register

Question ID : 4410091540311
Option 1 ID : 4410096084563
Option 2 ID : 4410096084566
Option 3 ID : 4410096084564
Option 4 ID : 4410096084565
Chosen Option : 2

Q.73 Let $C1 = (10, -4)$ and $C2 = (-6, 7)$
Using the same friend operator+ overload:
 $C3 = C1 + C2$
Find $C3$.

- Ans
- ☒ 1. 5, 4
 - ☒ 2. 4, 5
 - ☒ 3. 4, 3
 - ☒ 4. 3, 4

Question ID : 4410091540396
Option 1 ID : 4410096084894
Option 2 ID : 4410096084893
Option 3 ID : 4410096084891
Option 4 ID : 4410096084892
Chosen Option : 3

Q.74 Which of the following correctly identifies a key limitation of POP3 compared to IMAP?

- Ans
- ☒ 1. POP3 supports real-time synchronization of message state across devices, whereas IMAP does not.
 - ☒ 2. POP3 allows remote editing of email content on the server, whereas IMAP can only display messages locally.
 - ☒ 3. POP3 does not support server-side organization into folders and does not allow partial checking of message contents before download, whereas IMAP provides both capabilities.
 - ☒ 4. POP3 allows full folder management on the server, whereas IMAP requires downloading all messages before viewing their headers.

Question ID : 4410091535022
Option 1 ID : 4410096063328
Option 2 ID : 4410096063330
Option 3 ID : 4410096063327
Option 4 ID : 4410096063329
Chosen Option : 3

Q.75 Which of the following statements is TRUE regarding ordering in SQL relations?

- Ans
- ☒ 1. SQL guarantees that tuples will be stored and returned in sorted order unless explicitly modified by the user.
 - ☒ 2. Tuples in a relation are ordered according to their insertion sequence, but attributes remain unordered.
 - ☒ 3. Both attributes and tuples in a base table are inherently unordered because a relation is defined as a mathematical set.
 - ☒ 4. Attributes in a base table have a defined left-to-right order as specified in the CREATE TABLE statement, but tuples (rows) are not considered ordered in a relation.

Question ID : 4410091534913
Option 1 ID : 4410096062889
Option 2 ID : 4410096062887
Option 3 ID : 4410096062888
Option 4 ID : 4410096062890
Chosen Option : 4

Q.76 Which of the following statements is TRUE regarding First Fit and Best Fit?

- Ans ☒ 1. First Fit minimizes external fragmentation more effectively than Best Fit.
- ☒ 2. First Fit is faster but may lead to more fragmentation, whereas Best Fit reduces space wastage but requires more searching.
- ☒ 3. Both algorithms always produce identical memory layouts.
- ☒ 4. Best Fit always leads to faster allocation than First Fit.

Question ID : 4410091529899

Option 1 ID : 4410096042698

Option 2 ID : 4410096042700

Option 3 ID : 4410096042701

Option 4 ID : 4410096042699

Chosen Option : 2

Q.77 The fork() system call returns _____ if successful and _____ if unsuccessful.

- Ans ☒ 1. 0 if successful, 1 if unsuccessful
- ☒ 2. 1 to parent, 0 to child, and 2 if unsuccessful
- ☒ 3. Child PID to parent and 0 to child, and -1 if unsuccessful
- ☒ 4. Child PID to child, parent PID to parent, and 0 if unsuccessful

Question ID : 4410091529856

Option 1 ID : 4410096042521

Option 2 ID : 4410096042522

Option 3 ID : 4410096042520

Option 4 ID : 4410096042519

Chosen Option : 3

Q.78 In modular programming, function prototypes are usually stored in:

- Ans ☒ 1. The main function
- ☒ 2. .h header files
- ☒ 3. Preprocessor directives
- ☒ 4. .cpp files

Question ID : 4410091540419

Option 1 ID : 4410096084985

Option 2 ID : 4410096084984

Option 3 ID : 4410096084986

Option 4 ID : 4410096084983

Chosen Option : 3

Q.79 What happens if you declare a pointer but don't initialize it before dereferencing?

- Ans
- ☐ 1. It points to NULL automatically
 - ☐ 2. It points to the variable declared just before it
 - ☒ 3. It results in undefined behavior / "wild pointer"
 - ☐ 4. The compiler prevents dereferencing uninitialized pointers

Question ID : 4410091540354

Option 1 ID : 4410096084727

Option 2 ID : 4410096084729

Option 3 ID : 4410096084728

Option 4 ID : 4410096084730

Chosen Option : 3

Q.80 Which of the following is TRUE about IoT BIG data streams?

S1: They often arrive continuously from sensors
S2: They require real-time or near real-time analytics
S3: They always follow tabular structure
S4: They may contain noise and incomplete values

- Ans
- ☐ 1. Only S1 and S3
 - ☒ 2. Only S1, S2 and S4
 - ☐ 3. Only S1 and S2
 - ☐ 4. Only S3 and S4

Question ID : 4410091513014

Option 1 ID : 4410095976088

Option 2 ID : 4410095976087

Option 3 ID : 4410095976086

Option 4 ID : 4410095976089

Chosen Option : 2

Q.81 What is the primary purpose of memory partitioning in operating systems?

- Ans
- ☐ 1. To ensure processes share the same memory address
 - ☐ 2. To guarantee that processes are stored contiguously at all times
 - ☐ 3. To eliminate all overhead related to memory
 - ☒ 4. To divide main memory into manageable sections so multiple processes can be loaded

Question ID : 4410091529889

Option 1 ID : 4410096042659

Option 2 ID : 4410096042662

Option 3 ID : 4410096042661

Option 4 ID : 4410096042660

Chosen Option : 4

Q.82 Which of the following statements are TRUE about ARM processor?

- S1: It uses Load-Store architecture
S2: It belongs to CISC processor family
S3: It belongs to RISC processor family
S4: It does not support pipeline concept

- Ans ☐ 1. Only S3 and S4
☐ 2. Only S1, S2 and S4
☒ 3. Only S1 and S3
☐ 4. Only S1 and S2

Question ID : 4410091510258
Option 1 ID : 4410095965554
Option 2 ID : 4410095965555
Option 3 ID : 4410095965553
Option 4 ID : 4410095965552
Chosen Option : 4

Q.83 When adding a new attribute to an existing relation using ALTER TABLE, which of the following statements is TRUE if no DEFAULT clause is specified?

- Ans ☐ 1. The new attribute is automatically assigned system-generated values for all existing tuples.
☐ 2. The new attribute must be declared NOT NULL, and every existing tuple must immediately provide a value.
☒ 3. The new attribute will contain NULL in every existing tuple, and therefore a NOT NULL constraint cannot be applied while adding it.
☐ 4. The new attribute is added only to newly inserted tuples, not to the existing ones.

Question ID : 4410091534921
Option 1 ID : 4410096062920
Option 2 ID : 4410096062919
Option 3 ID : 4410096062922
Option 4 ID : 4410096062921
Chosen Option : 3

Q.84 Consider the following statements.

- S1: There is no single statement in C that can read, access, or print all elements of an array at once.
S2: To process all elements of an array, we must use a loop that repeatedly executes the same statement with different index values.

Which of the following is true?

- Ans ☐ 1. S1 is false, but S2 is true.
☒ 2. Both S1 and S2 are true.
☐ 3. S1 is true, but S2 is false.
☐ 4. Both S1 and S2 are false.

Question ID : 4410091529726
Option 1 ID : 4410096042000
Option 2 ID : 4410096042002
Option 3 ID : 4410096041999
Option 4 ID : 4410096042001
Chosen Option : 2

Q.85 If the function setval() assigns values beyond index size-1, what is the most likely consequence?

- Ans ☒ 1. Undefined behavior due to memory overrun
- ☐ 2. Compiler automatically prevents out-of-range assignments
- ☐ 3. Program automatically expands array
- ☐ 4. Memory leak

Question ID : 4410091540414
Option 1 ID : 4410096084966
Option 2 ID : 4410096084965
Option 3 ID : 4410096084964
Option 4 ID : 4410096084963
Chosen Option : 2

Q.86 Match the items in List A with their correct descriptions in List B related to the structure of an Inverted Page Table.

List A	List B
1. Page number	A. Identifies the process to which the virtual page belongs
2. Process identifier	B. Includes flags such as valid, referenced, modified, protection and locking information
3. Control bits	C. Contains the index of the next entry in case of a collision if no chain exists
4. Chain pointer	D. Part of the virtual address used to identify which page is referenced

- Ans ☐ 1. 1-A, 2-D, 3-C, 4-B
- ☒ 2. 1-D, 2-A, 3-B, 4-C
- ☐ 3. 1-B, 2-A, 3-D, 4-C
- ☐ 4. 1-C, 2-A, 3-B, 4-D

Question ID : 4410091529924
Option 1 ID : 4410096042800
Option 2 ID : 4410096042799
Option 3 ID : 4410096042798
Option 4 ID : 4410096042801
Chosen Option : 2

Q.87 Which of the following statements is TRUE about Von Neumann Architecture?

- Ans ☐ 1. It does not require control unit
- ☒ 2. It uses single bus for fetching code and data
- ☐ 3. Program and data are stored in separate memory
- ☐ 4. CPU has multiple buses for fetching instructions

Question ID : 4410091510244
Option 1 ID : 4410095965503
Option 2 ID : 4410095965501
Option 3 ID : 4410095965500
Option 4 ID : 4410095965502
Chosen Option : 2

Q.88 Consider the following statements.

Assertion (A):

In priority scheduling, starvation can occur when lower-priority processes wait indefinitely while higher-priority processes continue to execute.

Reason (R):

Aging is a technique that gradually increases the priority of waiting processes, helping prevent starvation.

Choose the correct option.

- Ans ☒ 1. Both A and R are true, and R is the correct explanation of A.
- ☐ 2. A is false, but R is true.
- ☐ 3. A is true, but R is false.
- ☐ 4. Both A and R are true, but R is not the correct explanation of A.

Question ID : 4410091529883

Option 1 ID : 4410096042635

Option 2 ID : 4410096042638

Option 3 ID : 4410096042637

Option 4 ID : 4410096042636

Chosen Option : 4

Q.89 Which of the following protocols is NOT used at the Network Layer of the TCP/IP model?

- Ans ☐ 1. ICMP (Internet Control Message Protocol)
- ☐ 2. IP (Internet Protocol)
- ☐ 3. ARP (Address Resolution Protocol)
- ☒ 4. TCP (Transmission Control Protocol)

Question ID : 4410091534974

Option 1 ID : 4410096063137

Option 2 ID : 4410096063138

Option 3 ID : 4410096063136

Option 4 ID : 4410096063135

Chosen Option : 4

Q.90 What is the main purpose of using a flag variable in the optimized bubble sort algorithm?

- Ans ☐ 1. To count the number of comparisons performed in each pass
- ☐ 2. To track the position of the smallest element
- ☒ 3. To detect if any swapping occurred in a pass and stop the algorithm early if the array is already sorted
- ☐ 4. To ensure that the inner loop executes only once per pass

Question ID : 4410091529840

Option 1 ID : 4410096042458

Option 2 ID : 4410096042457

Option 3 ID : 4410096042455

Option 4 ID : 4410096042456

Chosen Option : 3

Q.91 Which memory region holds the formal parameters of a function during execution?

- Ans
- ☒ 1. Code Segment
 - ☒ 2. Data Segment
 - ☒ 3. Stack
 - ☒ 4. Heap

Question ID : 4410091540341
Option 1 ID : 4410096084677
Option 2 ID : 4410096084676
Option 3 ID : 4410096084678
Option 4 ID : 4410096084675
Chosen Option : 1

Q.92 Which of the following best describes the role of a symbol table in a compiler?

- Ans
- ☒ 1. It lists lexical tokens scanned and removes them after the parsing process.
 - ☒ 2. It stores runtime variable values and manages them during program execution.
 - ☒ 3. It keeps records of names with attributes like type, scope, storage, and parameters for fast access.
 - ☒ 4. It supports linking by combining object files and resolving external symbols.

Question ID : 4410091539679
Option 1 ID : 4410096082050
Option 2 ID : 4410096082051
Option 3 ID : 4410096082052
Option 4 ID : 4410096082049
Chosen Option : 3

Q.93 Which of the following correctly matches the default CIDR prefixes for Class A, Class B, and Class C networks?

- Ans
- ☒ 1. Class A: /16, Class B: /24, Class C: /8
 - ☒ 2. Class A: /12, Class B: /17, Class C: /28
 - ☒ 3. Class A: /10, Class B: /20, Class C: /30
 - ☒ 4. Class A: /8, Class B: /16, Class C: /24

Question ID : 4410091534977
Option 1 ID : 4410096063148
Option 2 ID : 4410096063150
Option 3 ID : 4410096063149
Option 4 ID : 4410096063147
Chosen Option : 4

Q.94 A small computer system uses a 10-bit address bus, allowing it to access: $2^{10}=1024$ bytes. The designer wants to place 512 bytes of RAM in the lower addresses and 512 bytes of ROM in the upper addresses. Which address range will be assigned to the ROM?

- Ans
- ☐ 1. 000H – 1FFH
 - ☐ 2. 3FFH – 5FFH
 - ☐ 3. 100H – 2FFH
 - ☒ 4. 200H – 3FFH

Question ID : 4410091535062
Option 1 ID : 4410096063487
Option 2 ID : 4410096063490
Option 3 ID : 4410096063488
Option 4 ID : 4410096063489
Chosen Option : 4

Q.95 Under which condition can left recursion be safely eliminated from a syntax-directed translation scheme (SDT)?

- Ans
- ☐ 1. Left recursion cannot be eliminated from any SDT that performs semantic actions during parsing.
 - ☒ 2. Left recursion can be eliminated if the SDT contains only side-effect actions, or if attribute computations occur only in a postfix SDT.
 - ☐ 3. Left recursion can be eliminated only when the SDT uses inherited attributes and left-recursive rules.
 - ☐ 4. Left recursion can be eliminated only when the SDT contains no embedded semantic actions.

Question ID : 4410091539692
Option 1 ID : 4410096082104
Option 2 ID : 4410096082101
Option 3 ID : 4410096082102
Option 4 ID : 4410096082103
Chosen Option : 2

Q.96 RFID protocols in IoT operate primarily on which communication principle?

- Ans
- ☒ 1. Radio wave identification
 - ☐ 2. Wired serial communication
 - ☐ 3. Optical line-of-sight
 - ☐ 4. Infrared signal reflection

Question ID : 4410091512991
Option 1 ID : 4410095975995
Option 2 ID : 4410095975997
Option 3 ID : 4410095975994
Option 4 ID : 4410095975996
Chosen Option : 1

Q.97 Why does DNS use the Time-to-Live (TTL) field when caching resource-record mappings?

- Ans**
- ☐ 1. To allow recursive servers to permanently store mappings and avoid contacting authoritative servers again
 - ☒ 2. To ensure cached mappings expire after a specified time so outdated information is not returned to clients
 - ☐ 3. To force every DNS query to bypass caches and always reach the authoritative server
 - ☐ 4. To let caching servers modify the DNS records' validity period based on network load

Question ID : 4410091535003
Option 1 ID : 4410096063251
Option 2 ID : 4410096063252
Option 3 ID : 4410096063254
Option 4 ID : 4410096063253
Chosen Option : 2

Q.98 What does fopen("filename.txt", "r") return if the file does not exist?

- Ans**
- ☐ 1. A pointer to a new file
 - ☐ 2. An integer -1
 - ☒ 3. NULL
 - ☐ 4. A file pointer to an empty file

Question ID : 4410091540363
Option 1 ID : 4410096084766
Option 2 ID : 4410096084763
Option 3 ID : 4410096084765
Option 4 ID : 4410096084764
Chosen Option : 2

Q.99 In an IoT Publish-Subscribe communication model (such as MQTT), which of the following correctly describes the relationship between publishers and subscribers?

- Ans**
- ☐ 1. Must be on the same network
 - ☐ 2. Must know each other's IP addresses
 - ☐ 3. Must stay connected at the same time
 - ☒ 4. Are fully decoupled in time, space, and synchronization

Question ID : 4410091513001
Option 1 ID : 4410095976037
Option 2 ID : 4410095976034
Option 3 ID : 4410095976036
Option 4 ID : 4410095976035
Chosen Option : 4

Q.100 Which of the following statements correctly distinguishes regression from classification within supervised learning?

- Ans  1. Classification does not require labeled data, whereas regression always requires labeled data.
-  2. Regression is used to predict numeric quantities, while classification aims to predict categorical outcomes.
-  3. Regression predicts categories, while classification predicts continuous numeric values.
-  4. Both regression and classification are unsupervised learning techniques used without predefined labels.

Question ID : 4410091539734

Option 1 ID : 4410096082267

Option 2 ID : 4410096082265

Option 3 ID : 4410096082266

Option 4 ID : 4410096082268

Chosen Option : 2